AI Driven Wearable IoT systems for Cardiac and Stroke Risk Detection for Athletes

Submitted by

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Abstract

Strokes and heart issues among athletes are rare, but may lead to severe outcomes; identifying those at risk is difficult because intense physical conditioning alters normal body responses. This research proposes a system powered by artificial intelligence and connected wearable devices, designed to continuously monitor physiological signals and detect irregular heartbeat patterns associated with potential cardiac and stroke risks among athletes. Data drawn from multiple sources - tracking both bodily functions and athletic output - were analyzed using several learning methods: some based on probability trees, others relying on distance measures, linear separation, neural structures, or combined strategies, each weighed against speed and precision for live operation. Standing out within these trials was a mixed model pairing XGBoost with Random Forest logic, reaching 93.0 percent correct classifications while maintaining strong distinction between cases without demanding excessive processing power. Heart rate stood out in testing, along with measures tied to its variation, how intense movement was, force of motion, and rest patterns - each shaping predictions on heart health risks. Instead of relying solely on traditional methods, this system combines body worn sensors with smart algorithms that remain transparent in their logic. Scaling up becomes feasible when devices gather signals nonstop, feeding models capable of spotting irregular rhythms before symptoms arise. Personal risk estimates adjust over time, shaped by streams of fresh physiological inputs rather than one off assessments. Medical choices gain clarity as live analytics align with patient specific trends, quietly supporting doctors without replacing judgment. Prevention shifts forward not through grand claims but steady observation, especially useful where athletic performance meets long term wellness planning. Real world utility emerges slowly, built on consistency instead of sudden breakthroughs.This study is based on a physiologically informed synthetic dataset and should therefore be interpreted as a proof-of-concept rather than a clinically validated system.

Contents

List of Figures

Chapter 1

Introduction

1.1 Background of the Study

Competitive athletes are widely perceived as paradigms of cardiovascular health, yet they remain vulnerable to sudden cardiac events and stroke that, although rare, can be abrupt and devastating. Irregular heartbeats like atrial fibrillation raise the chance of blood clots leading to stroke; these might show up briefly during workouts or without symptoms, slipping past standard checks. Strange patterns in heartbeat and nervous system control often come before such incidents, offering a stretch of time - sometimes days, sometimes weeks - to catch problems early [10] [8] [5]. Still, current checkups rely too much on occasional doctor trips, calm-state readings, and broad risk formulas meant for average people, missing deeper clues about true heart condition. Not built for the intense heart changes, nervous system shifts, or unique physical needs seen in elite athletes. These tools struggle to catch fleeting irregular rhythms, unstable nerve control, yet show up during actual workouts or events. This results in a significant gap, as most assessments are performed under resting conditions while athletes experience substantial motion and physiological strain during training and competition. A major hole in how we track athlete well-being today [24] [19] [1] [23].

Advances in Internet of Things (IoT) technologies and Artificial Intelligence (AI) have recently made significant contributions toward overcoming this obstacle. Heart rate monitoring through IoT technology, alongside smartwatches and integrated fabric-based sensors, make it possible to collect highly detailed time-series information related to cardiac activity, autonomic nervous functions, and body motions during training sessions and competition periods [11]. Modern machine learning algorithms allow us to establish complex nonlinear connections among physiological parameters, environmental factors, and performance measures. The proposed solutions would provide a means of personalized risk analysis instead of relying on generic thresholds that apply to the whole population. The integration of wearable sensor technology with AI opens up the possibility of discovering early biomarkers, such as abnormalities in heart rate variability (HRV), or an imbalance in the autonomic nervous system, before the onset of a serious condition like an arrhythmia or stroke occurs. Based on this prospect, this thesis seeks to create an AI-enabled system for the early detection of risks associated with arrhythmia-related cardiovascular disease and stroke using wearable IoT devices based on HRV and other performance measures [25]. The overarching goal is not to provide definitive clinical diagnosis, but to deliver a scalable decision-support framework that continuously monitors athletes in their natural environments, identifies elevated cardiovascular risk at the earliest feasible stage, and generates timely alerts to guide further medical evaluation, training modification, and preventive intervention [23].

Stroke and Cardiovascular Risk: Stroke refers to the occurrence of neurologic symptoms that result from the loss of cerebral circulation either due to ischemic arterial occlusion or intracranial hemorrhage, although the former is significantly more common than the latter. Even though stroke is usually reported among patients who are at their geriatric age and have various clinical risks, strokes may be observed in young and apparently healthy individuals, such as athletes, especially those who have certain undiagnosed cardiovascular conditions. Research in cardiovascular risk assessment often relies on sample sizes that vary depending on the population studied, with typical ranges depending on specific health conditions and study design. The risk of serious cardiovascular outcomes varies across sports, but some adverse events may be reduced through early diagnosis, especially when arrhythmias are detected before severe complications occur. Cardiac arrhythmias, including atrial fibrillation, ventricular tachycardia, and atrioventricular block, are important cardiovascular conditions associated with increased stroke risk. As far as atrial fibrillation is concerned, this heart disease leads to blood pooling in the atrium and creates thrombi that can occlude cerebral arteries causing stroke. In view of this, early diagnostics of cardiovascular conditions predisposing to stroke is crucial. They fail to account for the adaptive changes, stress responses, and arrhythmias that may only appear during or following strenuous physical activity or that precede heart rhythm abnormalities, which are part of the autonomic changes [18]. This leads to the under diagnosis of arrhythmia-induced stroke risk, particularly in athletes who frequently subject their hearts to physical challenges through rigorous training.

Athletes as a Special Population: Athletes are considered an unusual physiological population because of the cardiovascular remodeling caused by training. Engaging for extended periods in high-intensity or high-volume workouts results in physiological changes, including stroke volume increase, vagal modulation, and morphological alterations in the heart chambers that are all collectively called ”the athlete’s heart.” These are usually not harmful but rather beneficial changes that, when assessed against standard references or diagnostic criteria for cardiovascular disease, may appear to mimic pathologies. Moreover, heart rate and blood pressure regulation in athletes are extremely variable depending on many factors, such as whether they are in the recovery, pre-exercise, or post-exercise phases, psychological state, and environmental variables[3]. On the contrary, current risk models for heart diseases are static and have been built on non-athletic groups. Sport-specific variables like the kind of discipline (endurance, power, and combined) are not accounted for, the distribution of the intensity of training, or the timing of the competitions. These assessments also fail to take into account the impact of training on what is considered normal with regard to cardiac structure and function. Using these models directly on athletes can be problematic because it can lead to either over- or underestimation of the risk. The former happens if the patient has exercise-induced or paroxysmal arrhythmias, while the latter can happen if benign changes from training are incorrectly interpreted as pathological findings[29] [28].

Role of Heart Rate Variability (HRV): Heart rate variability (HRV), intervals, is regarded as a powerful and noninvasive biomarker of autonomic nervous system function. Time domain, frequency domain, and non-linear analysis of HRV provide information regarding interactions between the sympathetic and parasympathetic systems on the level of sinoatrial node, while as autonomic nervous system function can be impacted by numerous physiological and pathological conditions, HRV measures are routinely used as markers of stress, fatigue, recovery capacity, and heart health [32]. The research suggests that the impairment of the autonomic system and/or decreased HRV is associated with the increased risk of malignant arrhythmia, sudden cardiac death, adverse outcomes after the onset of acute coronary and cerebrovascular conditions, and meta-analysis of studies of HRV-based atrial fibrillation detection found that sensitivities of the method were in the range of 94 percent and specificities exceeded 97 percent[36]. Moreover, advanced machine learning methods are able to attain area under curve values exceeding 0.95. On the other hand, HRV is used as a tool to monitor athletes' health and optimize their training programs to avoid negative consequences and promote adaptation, as HRV provides an opportunity to understand autonomic response to the cumulative load. Such characteristics make HRV an excellent candidate for early risk prediction in cases where autonomic dysfunction precedes the emergence of cardiac arrhythmia and stroke risks, since small changes in autonomic activity, captured through variations in HRV measures, could occur before arrhythmia or other clinical manifestations of such risks become evident. In doing so, HRV measures can be analyzed using machine learning algorithms to detect patterns associated with increased cardiovascular risk without relying on a simple thresholding approach. As HRV can be obtained continuously from wearable electrocardiogram (ECG) or photoplethysmography (PPG) data, it allows monitoring of the autonomic state in both resting and exercise conditions, which makes HRV fundamental for developing AI and wearables IoT applications in athletes [16].

IoT Wearables in Sports Health Monitoring: The availability of various wearable devices based on IoT technologies has revolutionized the process of physiological monitoring in sports and exercise training. Advanced chest bands, smart watches, and intelligent textiles with sensors embedded in them have the capacity to collect high-resolution electrocardiograms (ECGs), instant heart rate and HRV values, accelerometer and gyroscope readings, and supplementary measures like body temperature, blood oxygen saturation levels, and sleep quality in real time, which is communicated wirelessly to smart phones, edge computing systems, or cloud architectures for the purpose of providing continuous monitoring of athletes under normal training and competition circumstances [7]. Unlike clinical evaluations, which involve occasional measurements, wearable physiological sensors provide noninvasive and unobtrusive data collection, very high temporal resolution that makes it possible to detect sudden physiological changes, and contextual data that take into account real-world physical exertion, environment, and behaviors, thus offering new possibilities to discover early signs of cardiovascular stress and arrhythmias that may occur during particular types of training, recovery periods, and other conditions. The challenge lies in analyzing the resultant large, complex, and highly temporal data, which cannot be done using conventional rule- or threshold-based methods [33][12].

Need for Artificial Intelligence: The intricacies of and the sheer amount of physiological data from wearables require the application of artificial intelligence and machine learning techniques for its processing and early risk detection. Professional athletes continuously provide high-dimensional time-series data with non-linear dependencies, strong autocorrelations, and large inter-individual differences for many training sessions and competitions in terms of heart rate, HRV, motion, and context features [17]. Traditional approaches to analytics, which include threshold-based techniques, rule-based decision making, or simple univariate analyses, are inadequate for analyzing this data type because they are subject to false positives and false negatives, unable to detect complex multivariate dynamics that may indicate imminent risk of developing arrhythmia and increased stroke risk. The application of supervised learning algorithms, deep neural networks, or ensembles provides ways to extract feature embeddings from the data, integrate several different data types, and perform personal risk assessment [15]. Nonetheless, in order to deploy predictive models in sports scenarios, in addition to their accuracy and robustness, additional requirements regarding computation time must be satisfied due to inference being potentially performed on wearable devices and edge computing nodes. Consequently, there is a need for AI-based frameworks that balance model complexity with real-time performance, enabling continuous analysis of wearable data and timely generation of early warning alerts for arrhythmia-related cardiac and stroke risk in athletes[14].

1.2 Problem Statement

Despite extensive deployment of wearables in sports and the numerous advances in studying the use of HRV along with machine learning in detecting cardiovascular risks, there is no unified athlete-centered framework to address the problem of the early identification of the cardiovascular and stroke risks associated with arrhythmia. The current methods usually consider some isolated biomarkers such as the resting heart rate, HRV, etc., are generally designed for offline evaluation, and are mainly based on calibration using non-athlete patient samples. These methods also under-utilize the wealth of data available today thanks to wearable technology and also cannot detect any disturbances to the autonomic nervous system and the heart rhythm during the period preceding the event. Therefore, the current work aims to design and validate an Internet-of-Things and machine-learning-enabled framework for predicting the arrhythmia risk in multi-sport athletes. The main goal is to create a method to accurately predict the cardiovascular risk using HRV, physiological, and performance indicators obtained from wearable sensors, while being cognizant of the fact that final diagnosis and therapeutic intervention is done by a physician [21] [22] [34].

1.3 Research Objective

The primary objectives of this thesis are as follows:

- To create an IoT-based architecture for the ongoing collection and monitoring of HRV, physiological, and performance data from athletes through the use of wearable sensors.

- To extract and examine HRV, physiological, and activity-related variables that can be used to predict the development of autonomic dysfunction, arrhythmias, and increased risk of heart disease and strokes in athletes.

- To assess and compare machine learning algorithms for arrhythmia-based prediction of cardiovascular risks, taking into account not only the predictive performance of each algorithm but also the time required to make the predictions, which would allow the application of these algorithms in real-time scenarios.

- To determine the most effective physiological, autonomic, and performance indicators that can be used to predict cardiovascular and stroke risks among athletes.

Chapter 2

Related Work

2.1 Wearable IoT in Cardiovascular Monitoring

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, with sudden cardiac death (SCD) posing significant risk even in athletic populations [4]. Sudden cardiac death (SCD) is especially dangerous in athletes. Early diagnostics and constant physiological monitoring are key to mitigating the associated risks. However, classic ways of diagnosing the cardiovascular system are based mainly on episodic examinations like electrocardiogram (ECG), Holter monitoring, or routine physician appointments [20]. Wearable Internet of Things (IoT) technologies have revolutionized cardiovascular monitoring through constant and real-time collection of physiological data in non-clinical settings. Modern wearable IoT devices for CVD monitoring include patches for electrocardiographic (ECG) recordings, PPG-based smart watches, or heart-rate monitors worn around the chest [31] [13]. Huang et al. provides extensive analysis on more than 135 studies assessing the use of artificial intelligence in wearable sensor technology applied to cardiovascular diseases[11]. The authors discovered that ECG, PPG, and accelerometry sensors were predominant types. The clinical trials showed that sensitivity of atrial fibrillation using ECG patches was at 96.1%, while specificity was at 97.5%. On the other hand, using PPG-based smartwatches, the pooled sensitivity was at 97.4%, [7] while specificity was at 96.6%. However, these techniques have been shown to be sensitive to noise due to skin tone variations, physical motion artifacts, and incorrect placement especially during intense sporting events. While the commercial wearables are important in tracking the health of users, the majority have been shown to track either physical fitness or cardiac issues independently without AI prediction of athlete health status [35] [26].

2.2 Machine Learning for Cardiovascular Risk Prediction

ML methods have been shown to perform better than conventional risk-scoring methodologies in epidemiology when predicting risk factors for heart diseases. Unlike traditional risk scoring models, ML techniques are able to model nonlinear and multidimensional associations between physiologic variables. Shah et al. proposed a hybrid ensemble framework combining Gradient Boosting, CatBoost, LightGBM, Hybrid XGBoost, and Neural Networks, achieving an AUC-ROC of 0.82 with precision of 81%, recall of 83%, and F1-score of 82%. Comparative studies report Random Forest classification accuracies ranging between 88.7% and 90.78% across multiple cardiovascular datasets[25]. Deep learning models further augment prediction power, especially when used with unprocessed ECG signals. Models combining convolutional neural networks (CNNs) and long short-term memory networks (LSTMs) have been observed to achieve an accuracy of 86–99% for detecting arrhythmia. Furthermore, deep learning models based on HRV analysis achieve sensitivity and specificity values of 94% and 97%, respectively, for atrial fibrillation [30]. Mandal et al. showed that stress detection from heart rate data alone achieved 73% accuracy using Histogram-based Gradient Boosting, illustrating that even simplified physiological features can yield meaningful predictive insights without complex ECG feature engineering [18]. Despite strong predictive performance, many studies emphasize accuracy metrics without evaluating computational efficiency, limiting their applicability to real-time wearable environments[2] [9].

2.3 Multi-Modal Sensor Fusion in Healthcare

However, new technologies emphasize the importance of multi-modal sensor fusion whereby physiological, environmental, and biomechanical data are merged to improve predictive accuracy. For example, Villegas-Ch et al. created a combined temperature, humidity, and accelerometer-based model and successfully achieved anomaly detection with 98.7% accuracy. Most importantly, over a 12-week continuous learning process through feedback, accuracy increased linearly from 92% to 99%. Kajornkasirat et al. used a COVID-19 monitoring system which incorporated body temperature, heart rate, and SpO_2 data along with AI-based facial recognition of emotional states and obtained 80% classification accuracy via Random Forests algorithms [14]. Moreover, in sports science, inertial measurement units are popular tools in biomechanics and injury prediction. Souaifi et al. surveyed 73 research articles on AI applications in the field of sports biomechanics and reported that IMU-based systems identify neuromuscular fatigue with 87% sensitivity and predict the risks of hamstring injuries with 85% accuracy [27]. Furthermore, computer vision systems showed sub-centimeter accuracy compared to marker-based motion capture systems. Although the aforementioned literature proves that integrated IoT health monitoring is a viable concept, in most cases, cardiovascular risks, biomechanical injury risks, and training loads are considered independently.

2.4 AI in Clinical Applications

Interpretability of models remains a key issue that needs to be addressed for the deployment of such systems in healthcare settings. Interpretability is essential for healthcare providers in validating decision-making processes and guaranteeing patient safety. SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) techniques have been extensively used to improve transparency. For instance, Shah et al. adopted the SHAP method to explain the ensemble model developed in their study. Systolic/diastolic blood pressure, BMI, and cholesterol-glucose ratio were identified as the most prominent features influencing predictions made by the model. Intrinsically, attention mechanisms in DNNs offer interpretability by focusing on certain temporal intervals or inputs that significantly affect predictions. However, recent studies show that there exists a trade-off between the level of interpretability and peak performance of models, especially in complex architectures. For wearable-based cardiovascular disease monitoring systems, interpretability and computational efficiency need to be reconciled[6].

2.5 Research Gaps

Despite significant progress in wearable IoT systems and machine learning-based cardiovascular risk prediction, several limitations remain:

- Limited Real-World Validation: Many studies are conducted in controlled laboratory environments, reducing generalizability to real-world athletic conditions where motion artifacts and environmental variability are prevalent.
- Fragmented Monitoring Approaches: Existing systems often treat cardiovascular risk, training load, and biomechanical monitoring independently rather than as integrated physiological processes.
- Insufficient Computational Benchmarking: Most prior research emphasizes predictive

accuracy while neglecting systematic evaluation of computational performance, including training time, inference latency, memory usage, and model size.

- Deployment Feasibility Gaps: Few studies analyze algorithm suitability for real-time wearable or edge deployment where latency, battery consumption, and storage constraints are critical.

These gaps highlight the need for comprehensive evaluation frameworks that jointly assess predictive accuracy and computational efficiency to support practical, real-time monitoring of athlete health.

Chapter 3

Methodology

3.1 Dataset and Data Acquisition

A thorough dataset of a synthetic athlete's health monitoring was constructed to provide an emulation of physiological and biomechanical parameters measured using multi-modal IoT wearable devices during exercise training sessions. In total, 10,000 time records were generated based on 5 minutes sampling periods during which the physiological parameters were simulated continuously. The synthetic data set was used to evaluate the performance of the proposed approach while maintaining ecological validity. This was achieved by ensuring that the physiological parameter distribution aligned with existing research in athletic physiology. The dataset encompassed comprehensive athlete profiles spanning ages 18-85 years

TOP 20 HIGHEST RISK ATHLETES:

ID	Age	Risk Level	Risk Score	Avg HR	BP	Avg SpO2	Arrhythmia	Stress
S001	59	CRITICAL	7	83.2	121/97	97.5	14	2.1
S002	50	CRITICAL	7	80.2	123/104	97.6	5	2.1
S003	28	CRITICAL	7	80.8	137/83	97.9	6	2.2
S006	26	CRITICAL	7	80.2	169/60	97.9	7	2.3
S036	46	CRITICAL	7	80.2	98/75	97.8	9	2.3
S007	54	CRITICAL	7	83.2	118/72	97.8	11	2.3
S008	79	CRITICAL	7	80.3	130/60	98.0	5	2.0
S009	60	CRITICAL	7	81.1	91/97	97.8	6	2.3
S012	59	CRITICAL	7	78.9	90/82	97.8	9	2.0
S010	66	CRITICAL	7	79.0	133/83	98.1	9	2.1
S014	56	CRITICAL	7	77.7	129/66	97.6	5	1.8
S013	34	CRITICAL	7	82.7	129/77	97.8	10	2.2
S017	64	CRITICAL	7	77.4	106/81	98.2	6	1.8
S018	36	CRITICAL	7	78.4	129/78	97.9	7	1.9
S015	54	CRITICAL	7	80.6	90/112	97.9	7	2.0
S016	49	CRITICAL	7	81.8	123/82	97.8	6	1.9
S019	42	CRITICAL	7	80.8	133/62	98.0	8	2.1
S020	46	CRITICAL	7	79.6	129/60	97.7	8	2.0
S023	19	CRITICAL	7	81.7	124/60	97.9	4	2.5
S022	48	CRITICAL	7	82.2	111/73	97.9	4	2.0

Figure 3.1: Comprehensive Physiological Analysis and Risk (All Athelete)

(μ = 44.7, σ = 14.5 years), with gender distribution across male and female athletes, and a rich multivariate feature space capturing physiological, biomechanical, and psychophysiological parameters.

Figure 3.1 shows a detailed physiological risk analysis of the top 20 highest-risk athletes in the dataset using important health indicators such as age, risk score, average heart rate, blood pressure, oxygen saturation (SpO), arrhythmia count, and stress level. This figure is important because it shows how different health parameters can be combined to identify athletes who may have higher cardiovascular risk. It also supports the usefulness of the proposed AI-based monitoring system for early detection, better health management, and improved athlete safety.

Cardiovascular measurements include baseline heart rate (45–183 beats per minute), systolic blood pressure (90–196 mm Hg), diastolic blood pressure (60–120 mm Hg), mean arterial pressure calculated using the following formula:

$$\text{MAP} = \frac{1}{3}\text{SBP} + \frac{2}{3}\text{DBP} \tag{3.1}$$

and pulse pressure. The respiratory markers include respiratory frequency (8.8–35 breaths per minute) and oxygen saturation (SpO_2: 90.4–100.0%).Thermoregulatory measures include core body temperature (35.5–41.1°C), environmental temperature (10–45°C), relative humidity (20–95%), and calculated heat index.HRV indicators, which are considered reliable indices of autonomic nervous system activity and cardiovascular risk, include RMSSD (root mean square of successive differences: 0–47.4 ms), pNN50 (percentage of NN intervals differing from preceding one by ¿50 ms: 0–47.4%), SDNN (standard deviation of NN intervals: 20–102.8 ms), and triangular index (TI: 8–45.6).Biomechanical and activity measurements include daily number of steps (2,814–15,363), intensity of physical activity levels (sedentary, light, moderate, vigorous activities), and three-axis IMU data that comprise acceleration measurements ($\text{accel}_{x/y/z}$: −2.0–2.5 g) and gyroscope measurements ($\text{gyro}_{x/y/z}$: −10).

Arrhythmia detection was the target feature to be classified using a primary classifier (binary target classification) defined as follows: 0 – No Arrhythmia and 1 – Arrhythmia Detected; secondarily, it was classified based on health risks defined as Normal, Elevated, and Critical. The provided data set had real-life class imbalance commonly seen in cardiovascular monitoring applications because arrhythmia occurred in 1,141 out of 10,000 readings (11.4%).

3.2 Data Preprocessing and Feature Engineering

3.2.1 Data Quality Assessment and Cleaning

The initial exploration of the data revealed that there were no missing observations in any of the 32 variables analyzed; therefore, the problem of handling missing data was not considered in this study. The descriptive statistics (i.e., mean, standard deviation, quartiles, minima, maxima) of all continuous variables were calculated to detect any outliers or physiologically impossible values.

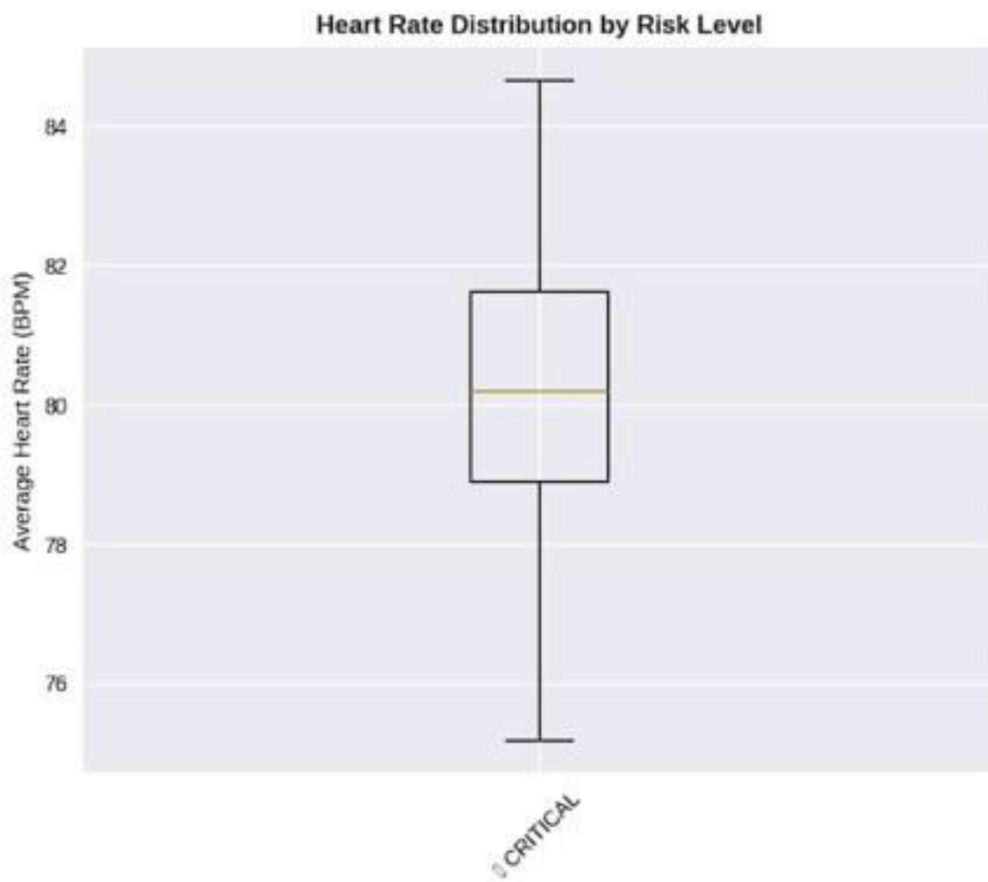

Figure 3.2: Activity Level Distribution by Risk Level

Figure 3.2 presents the distribution of activity level, measured by step count, for athletes classified under the critical risk category using a boxplot. The figure shows the median, spread, and variation in daily steps among high-risk athletes, indicating differences in physical activity behavior within this group. This figure is important because it helps examine the relationship between activity patterns and health risk, supporting the use of movement-related data as a valuable feature in the proposed AI-based monitoring system.

3.2.2 Feature Engineering and Derived Metrics

Beyond raw sensor readings, engineered features were computed to enhance the feature space and improve model discriminability. These included:

- **Magnitude of acceleration:** Euclidean norm of 3-axis accelerometer data, computed as $M_{acc} = \sqrt{accel_x^2 + accel_y^2 + accel_z^2}$ quantifying overall movement intensity independent of direction.
- **Magnitude of angular velocity:** Euclidean norm of 3-axis gyroscope data, computed as $M_{acc} = \sqrt{accel_x^2 + accel_y^2 + accel_z^2}$, capturing rotational motion during athletic activities.

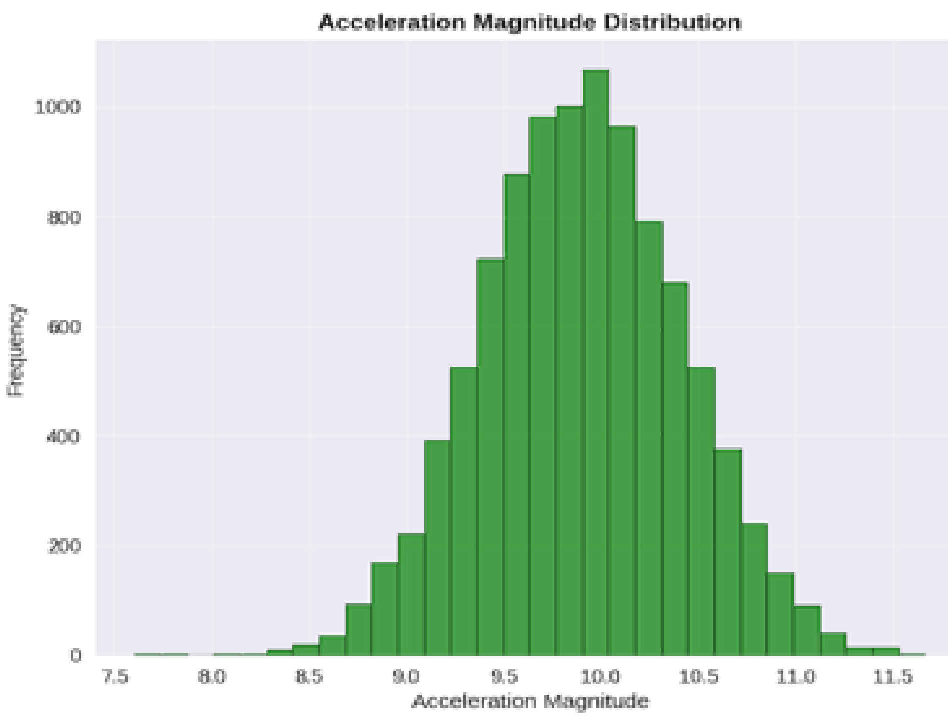

Figure 3.3: Acceleration Magnitude Distribution Metrics

Figure 3.3 shows the distribution of acceleration magnitude values obtained from athlete movement data using a histogram. The figure indicates that most acceleration values are concentrated around the central range, while fewer observations occur at lower and higher extremes. This figure is important because it helps understand movement intensity patterns and variability, supporting the use of acceleration-based features in the proposed AI system for monitoring physical activity and identifying potential health risks.

- **Composite activity intensity:** Normalized composite metric integrating activity level categories and accelerometer magnitude, scaled to [0, 1] interval to represent overall physical exertion.

- **SpO_2 deviation from baseline:** Computed as $\Delta SpO_2 = 100 - SpO_{2,measured}$, quantifying the magnitude of hypoxemia relative to normal saturation.

- **Mean arterial pressure:** Calculated as equation 3.1, a more physiologically relevant pressure metric than individual systolic or diastolic measurements.

3.2.3 Data Normalization and Standardization

All continuous features were standardized using z-score normalization (Standard Scaler in scikit-learn), transforming each feature x to a standardized score $z = \frac{x-\mu}{\sigma}$, where μ is the sample mean and σ is the standard deviation. This normalization procedure ensures zero mean and unit variance across all features, preventing scale-dependent bias in distance-based algorithms (KNN, SVM) and facilitating numerical stability and faster convergence in gradient-based learning algorithms (neural networks, gradient boosting).

3.2.4 Class Imbalance Mitigation

To address the inherent class imbalance (88.6% negative vs. 11.4% positive cases), a hybrid resampling strategy was applied prior to model training:

- **Synthetic Minority Over-sampling Technique (SMOTE):** Sampling strategy = 0.5 to synthetically generate 1,500 additional minority-class samples via k-nearest neighbors interpolation ($k = 5$), bringing the minority class to 50% of the majority class size.

- **Random under-sampling:** Majority class reduced by factor of 0.6 to achieve final balanced distribution approaching 1:1 ratio, thereby eliminating algorithmic bias toward the negative class.

- **Stratified sampling:** Maintained class distribution across all subsequent train/validation/test partitions.

3.2.5 Train-Validation-Test Stratification

The complete dataset of 10,000 samples was partitioned using stratified random sampling (random_state=42 for reproducibility) to preserve class proportions across splits:

- **Training set:** 70% of data (7,000 samples), used for model estimation and hyperparameter tuning.
- **Validation set:** 15% of data (1,500 samples), reserved for hyperparameter selection and early stopping.
- **Test set:** 15% of data (1,500 samples), held out until final model evaluation to provide unbiased performance estimates.

Stratification ensured that positive and negative classes maintained their original proportions within each partition, preventing evaluation bias that could arise from imbalanced random splits.

3.3 Machine Learning Model Selection and Architectures

The study investigated a set of eleven different machine learning algorithms representing varying architectures and complexities, including statistical models, decision trees, distance-based models, and deep neural networks. Using multiple algorithms from various perspectives enabled accurate comparisons between the models and their effectiveness.

3.3.1 Base Learner Algorithms

- **Random Forest:** An ensemble method consisting of 100 decision trees trained on bootstrap samples of the training data. Hyperparameters: n_estimators=100, max_depth = None (unlimited tree depth), min_samples split=2, class weight='balanced' (automatic adjustment for class imbalance via inverse frequency weighting), random_state=42. Random Forest achieves robustness through variance reduction via bootstrap aggregation and feature randomization at each split.
- **Gradient Boosting Machine (GBM):** A sequential ensemble method building 100 weak learners (trees) that iteratively correct predecessor errors. Configuration: n_estimators=100, learning_rate=0.1 (shrinkage parameter for regularization), max_depth = 3 (shallow trees), subsample=0.8 (stochastic gradient boosting), random_state = 42. GBM prioritizes reduction of prediction bias through additive corrections.
- **Hybrid XGBoost (Extreme Gradient Boosting):** Regularized gradient boosting with parallel tree construction and second-order gradient information. Hyperparameters: n_estimators = 100, learning_rate = 0.1, max depth = 6, subsample=0.8 (row

sampling), colsample_bytree = 0.8 (column sampling),scale_pos_weight (computed as ratio of negative to positive samples for class imbalance handling), random_state=42. Hybrid XGBoost's computational efficiency stems from histogram-based learning and approximate split finding.

- **LightGBM (Light Gradient Boosting Machine):** Gradient-based learning exploiting leaf-wise tree growth and gradient-based one-side sampling (GOSS). Configuration: n_estimators = 100, learning rate = 0.1, max_depth = -1 (unrestricted), num_leaves = 31, class_weight = 'balanced', random_state = 42. LightGBM achieves superior memory efficiency and training speed through histogram quantization.

- **CatBoost (Categorical Boosting):** Boosting framework with built-in categorical feature handling via target encoding. Hyperparameters: iterations=100, learning rate = 0.1, depth = 6, auto_class_weights = 'Balanced', random_state = 42, verbose = 0. CatBoost employs ordered boosting to reduce overfitting bias in gradient estimation.

- **Support Vector Machine (SVM):** Non-parametric discriminative classifier using radial basis function (RBF) kernel for nonlinear decision boundaries. Specifications: kernel='rbf', C = 1.0 (regularization parameter), gamma = 'scale' (inverse of number of features), class_weight = 'balanced', probability = True (output Platt-scaled probabilities), random_state = 42. SVM finds maximum-margin hyperplane in transformed feature space.

- **K-Nearest Neighbors (KNN):** Instance-based lazy learner classifying samples based on majority class of k=5 nearest neighbors. Hyperparameters:n_neighbors = 5, weights = 'distance' (inverse distance weighting), metric = 'euclidean', algorithm = 'auto'(automatic selection of optimal implementation). KNN provides non-parametric local approximation.

- **Logistic Regression:** Linear probabilistic classifier with L2 regularization. Configuration: penalty = 'l2', C = 1.0 (inverse regularization strength), solver = 'lbfgs' (quasi-Newton optimization), class_weight = 'balanced', max_iter = 1000, random_state = 42. Logistic regression models posterior class probability via sigmoid function.

- **Gaussian Naïve Bayes:** Probabilistic classifier assuming conditional independence of features given class membership and Gaussian feature distributions. Default parameters employed; no hyperparameter tuning applied. Naïve Bayes provides fast, parameter-free baseline.

- **Decision Tree:** Single decision tree with depth constraints to prevent overfitting. Hyperparameters: max depth = 5 (maximum tree depth), min_samples split = 10 (minimum samples required to split node), min_samples_leaf = 5 (minimum samples in leaf nodes), class_weight='balanced', random state = 42. Decision trees provide inherent interpretability through transparent decision rules.

3.3.2 Hybrid Ensemble Architectures

Beyond individual algorithms, three hybrid ensemble models were developed using stacked generalization (stacking), a meta-learning approach wherein base learner predictions serve as inputs to a secondary meta-learner. Stacking architecture comprises:

- **Base learners (Level 0):** Five diverse algorithms- Random Forest, Gradient Boosting, SVM, KNN, and Neural Network trained on original training data. Their predictions on validation folds are used to construct meta-features.
- **Meta-learner (Level 1):** A secondary algorithm trained on base learner predictions. Three meta-learner variants were evaluated:
  - Hybrid-Hybrid XGBoost: Meta-learner = Hybrid XGBoost (same hyperparameters as base Hybrid XGBoost)
  - Hybrid-LightGBM: Meta-learner = LightGBM (same hyperparameters as base LightGBM)
  - Hybrid-CatBoost: Meta-learner = CatBoost (same hyperparameters as base CatBoost)

Stacking was implemented via five-fold cross-validation on the training set. For each of the five folds, base learners were trained on four folds and produced predictions on the held-out fold, generating out-of-fold (OOF) predictions. These OOF predictions from all folds were concatenated to create meta-training features. The meta-learner was then trained on these meta-features with corresponding target labels, preventing data leakage and reducing overfitting risk.

3.3.3 Deep Neural Network Architecture

A fully connected feedforward neural network was developed with explicit regularization to prevent overfitting:

Network Architecture:

- Input layer: 18 features (after feature selection and preprocessing)
- Hidden layer 1: 128 neurons, ReLU activation function
- Dropout layer 1: Dropout rate = 0.3 (30% of units randomly deactivated during training)
- Hidden layer 2: 64 neurons, ReLU activation function
- Dropout layer 2: Dropout rate = 0.3
- Output layer: 2 neurons (binary classification), Softmax activation function

Training Configuration:

- Optimizer: Adam (adaptive learning rate = 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-7}$)
- Loss function: Categorical cross-entropy
- Batch size: 32 samples per gradient update
- Epochs: 50 maximum, with early stopping (patience=10 epochs without validation loss improvement)
- Validation split: 20% of training data reserved for validation monitoring
- Metrics: Categorical accuracy

The dropout layers provided L2-like regularization by stochastically removing activations, reducing co-adaptation and improving generalization. ReLU activation functions introduced nonlinearity, and the Adam optimizer adapted learning rates per parameter for efficient convergence.

3.3.4 Model Training and Cross-Validation

Each model was trained on the 7,000-example training set with their own sets of hyperparameters, which are defined in Section 3.3. Initial hyperparameter values were selected according to machine learning guidelines and previous work related to cardiovascular risk prediction. Fixed hyperparameters were used for each experiment so that a direct computation efficiency comparison could be made independent of any hyperparameter optimization time.

K-Fold Cross-Validation: The five-fold stratified cross-validation was conducted on the training data to test the generalizability and robustness of the model. Five-fold cross-validation means that the training dataset was split into five parts, and then the model was trained on four folds out of five (80% of training data). Afterward, the model was tested using the remaining fold, and this process repeated five times until all five folds were used as testing data.

Computational Environment: All experiments were conducted on identical hardware under controlled conditions to ensure reproducible timing measurements:

- Processor: Intel Core i7 (8th generation or equivalent)
- RAM: 16 GB DDR4 SDRAM
- Storage: SSD (minimal I/O latency)
- GPU: None (CPU-only computation)
- Operating System: Linux (Ubuntu 20.04 LTS)
- Python version: 3.9.x
- Key libraries: scikit-learn (1.0.2), Hybrid XGBoost (1.6.1), LightGBM (3.3.2), CatBoost (1.0.6), TensorFlow (2.8.0), NumPy (1.21.0), Pandas (1.3.5)

Maintaining identical hardware and software configurations across all model evaluations ensured that timing measurements reflected algorithmic efficiency rather than environmental variations.

3.3.5 Performance Evaluation Framework

Model assessment encompassed two complementary dimensions: (1) predictive accuracy, quantifying discriminative ability for cardiovascular risk identification, and (2) computational efficiency, assessing real-time deployment feasibility on resource-constrained wearable platforms.

Predictive Performance Metrics

On the held-out test set (1,500 samples), the following classification metrics were computed:

- **Accuracy:** The proportion of correct predictions among all predictions, computed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.2}$$

where TP (true positives), TN (true negatives), FP (false positives), and FN (false negatives) are derived from the confusion matrix.

- **Precision (Positive Predictive Value):** The fraction of positive predictions that are correct, reflecting false alarm rate:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.3}$$

- **Recall (Sensitivity, True Positive Rate):** The fraction of actual positive samples correctly identified, reflecting detection completeness:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3.4}$$

- **F1-Score:** The harmonic mean of precision and recall, providing a balanced performance metric when precision and recall trade-offs exist:

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \tag{3.5}$$

- **Area Under the Receiver Operating Characteristic Curve (AUC-ROC):** The AUC indicates the model's performance in discriminating positive samples from negative ones at all levels of classification. It is calculated using numerical integration on the ROC curve. The AUC ranges between [0, 1]; a value of 0.5 implies randomness while 1 implies perfect discrimination.

- **Confusion Matrix Analysis:** This analysis was done for the value of diagnosing from the 2×2 matrix using TN (true negatives), FP (false positives), FN (false negatives), and TP (true positives). With FN (missed cases of arrhythmias) especially expensive in clinical practice, recall must be analyzed, too.

- **Cross-Validation Statistics:** Five-fold CV mean accuracy and standard deviation (CV Mean ± CV Std) provided measures of model stability and robustness across different training data samples.

Computational Efficiency Metrics (Primary Contribution)

To address the critical gap in computational performance assessment and enable practical algorithm selection for real-time wearable deployment, the following efficiency metrics were systematically measured and reported:

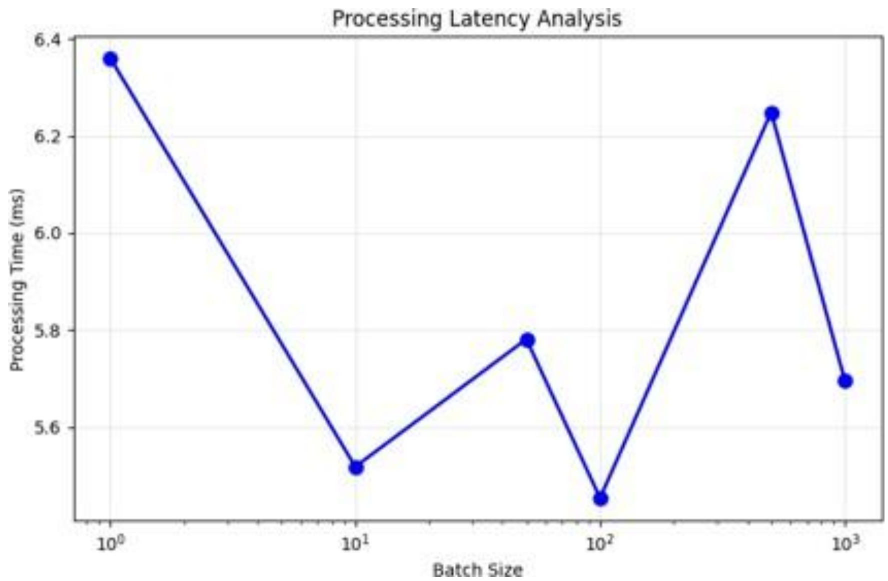

Figure 3.4: Processing Latency Analysis

Figure 3.4 presents the processing latency analysis of the proposed system across different batch sizes. The graph shows how the prediction time changes as the number of processed samples increases, providing insight into the computational speed and response efficiency of the model. This figure is important because low and stable latency is essential for real-time wearable applications, where fast decision-making is required for continuous athlete health monitoring.

Batch Processing Performance Analysis

- **Training Time (seconds):** Total wall-clock elapsed time from initiating model training (`.fit()` method) to completion of training procedure. Measured using Python's `time.time()` function with microsecond precision. All base learners and hybrid meta-learners were trained once; training time represents the single training execution on the full training set. For neural networks, training time includes all 50 epochs (or early stopping termination).
- **Per-Sample Inference Latency (milliseconds per sample):** The prediction latency for a single sample, computed as the ratio of total test set prediction time to test

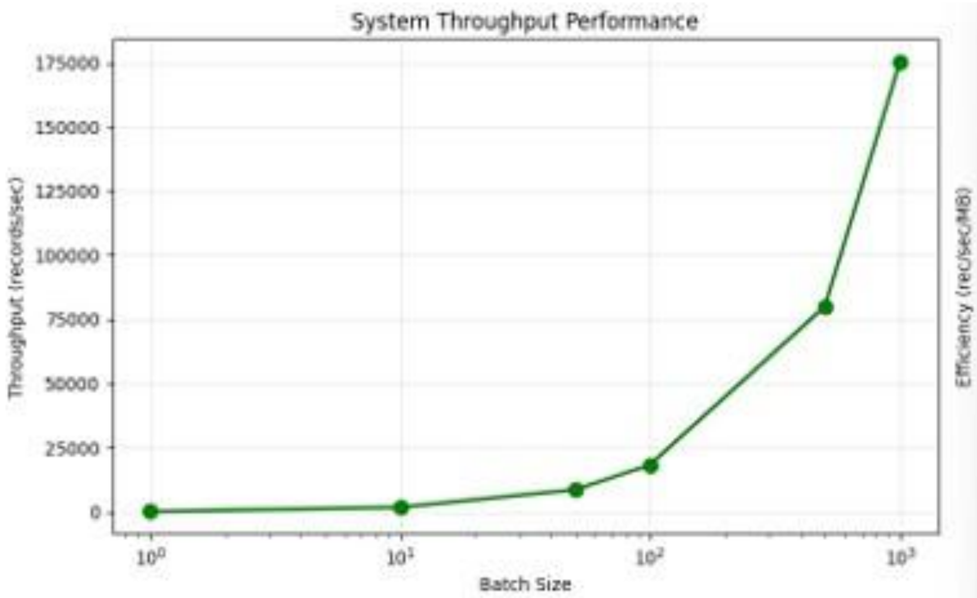

Figure 3.5: System Throughput Performance

set size:

$$\text{Latency}_{\text{per sample}} = \frac{\text{Total prediction time for 1500 test samples (ms)}}{\text{1500 samples}} \tag{3.6}$$

Latency measurement consisted of recording wall-clock time before and after calling .predict() with the entire test set, followed by division by the number of samples. Latency is an indication of the time taken for making a single prediction, which in real-time wearable applications needs to be under 100 milliseconds.

Figure 3.5 presents the system throughput performance across different batch sizes, showing how many samples can be processed within a given time. The graph indicates that throughput increases significantly as batch size becomes larger, demonstrating improved computational efficiency. This figure is important because high throughput is essential for scalable real-time wearable monitoring systems, where continuous physiological data from many users must be processed quickly and reliably.

- **System Throughput (samples per second):** The number of samples processed per unit time, providing the inverse perspective:

$$\text{Throughput} = \frac{1}{\text{Latency}_{\text{per sample (seconds)}}} \tag{3.7}$$

Throughput quantifies batch processing capacity and indicates how many simultaneous athlete streams a single model instance could service concurrently.

- **Model Size (megabytes):** The serialized model file size after training, stored using joblib format for scikit-learn models or native binary formats for Hybrid XGBoost, LightGBM, CatBoost, and TensorFlow models. Model size directly impacts storage requirements on edge devices and download bandwidth for over-the-air model updates in deployed wearable systems.
- **Peak Memory Usage (megabytes):** Maximum RAM consumption during model training and inference, measured via the memory_profiler library. Memory profiling recorded peak working set size at granular temporal resolution, capturing the maximum memory footprint across all phases of model execution.

All timing and resource measurements were conducted under controlled conditions with minimal background processes, hardware monitoring disabled, and CPU frequency scaling disabled (fixed to maximum frequency) to minimize environmental noise in timing measurements.

Scalability was assessed based on inference latency and throughput at different batch sizes, such as 1, 10, 50, 100, 500, and 1000 for batch-processing applications, such as batch processing end-of-day athlete data collected through distributed sensor devices. The latencies and memory usage associated with each batch size were documented to understand scalability properties.

3.3.6 Feature Importance Analysis

For tree-based ensemble models (Random Forest, Gradient Boosting, Hybrid XGBoost, LightGBM, CatBoost, Decision Tree), feature importance scores were extracted post-hoc to identify the most discriminative variables for arrhythmia detection:

- **Gini Importance (Mean Decrease in Impurity):** For decision tree-based models, importance scores reflect the weighted average decrease in impurity (Gini index or entropy) at each split, summed across all trees in the ensemble:

$$I_j = \frac{1}{T} \sum_{t=1}^{T} \left(\text{Impurity}_{\text{parent}} - \text{Impurity}_{\text{children}} \right) \tag{3.8}$$

  where j is the feature, T is the total number of trees, and the impurity metrics measure Gini index decrease at each node.

- **Gain-Based Importance:** For boosting methods (Hybrid XGBoost, LightGBM, CatBoost), importance is computed as the total gain reduction across all splits using the feature:

$$I_j = \sum_{\text{splits using } j} \text{Gain}_{\text{split}} \tag{3.9}$$

  where Gain represents the loss reduction from the split.

The top 15 features were ranked and visualized to guide clinical interpretation and inform feature selection for potential lightweight deployment models targeting edge devices with severe computational constraints.

3.3.7 Visualization and Reporting

Comprehensive visualizations were systematically generated to facilitate multi-dimensional model comparison, clinical interpretation, and communication of findings. Although the dataset was designed to reflect realistic physiological patterns, it does not capture real-world variability such as sensor noise, motion artefacts, and inter-individual differences. Therefore, the results of this study represent an initial methodological validation and require further testing on real-world athlete data before practical deployment.

- **Classification Performance Visualizations:**
  - Confusion matrices of all the 11 models were presented in a 4x3 table format that allows for easy comparison of false negatives (sensitivity) and false positives (specificity).
  - ROC curves of all models plotted in a single figure, showing their AUC values.
- **Efficiency-Accuracy Trade-offs:**
  - Graph plotting time spent on training (x-axis) against accuracy on the test dataset (y-axis), showing the tradeoff between accuracy and computational efficiency for each model and labeled with the corresponding model names.
  - Scatter plot of processing delay vs. accuracy.
- **Feature Importance Analysis:** Horizontal bar chart ranking top-15 features by importance score from Random Forest model, displaying feature coefficients with color-coded gradient.
- **Physiological Data Distributions:**

- Complete dashboard of univariate and bivariate distributions categorized based on the presence/absence of arrhythmia:
  * Body temperature distribution with normal values highlighted
  * Scatter plot of blood pressure (diastolic vs. systolic) with hypertension ranges
  * Distribution of SpO levels with hypoxemia cutoff (95%)
  * Heart rate distribution based on arrhythmia
  * Distributions of HRV parameters, showing autonomic nervous system differentiation
  * Scatter plot of stress vs. heart rate
  * Distribution of sleep quality
  * Histogram for respiration rate
  * Distribution of activity level categorized by risk

- **Real-Time Processing Performance:**
  - Plot showing latency (in ms) versus batch size for all models
  - Throughput (in samples/second) versus batch size
  - Consumption of memory versus batch size
  - Efficiency of processing (in samples/joules, via throughput and power consumption estimates)

All plots have been created with Matplotlib (version 3.5.0) and Seaborn (version 0.11.2) at high quality (dpi=300; vector-based output).

3.3.8 Statistical Analysis and Significance Testing

Given the highly consistent accuracy across top-performing models (92.5–93.0%, range: 85.7–92.95%), traditional statistical significance testing (e.g., McNemar's test, paired t-tests) would likely detect negligible practical differences. Accordingly, statistical analysis focused on:

- **Cross-validation stability assessment:** Coefficient of variation (CV Std / CV Mean) quantified model consistency across 5 folds; lower CV indicated more stable generalization.

- **Confidence interval construction:** 95% confidence intervals were computed for test set accuracy via binomial proportion confidence intervals using the Wilson score method, accounting for finite sample size ($n = 1,500$).

- **Effect size interpretation:** Rather than p-values, absolute differences in metrics (AUC difference of 0.01 units, latency differences of 0.01 ms) were interpreted using domain-specific standards for wearable and clinical applications.
- **Computational efficiency as primary discriminator:** Given equivalent predictive accuracy, training time and inference latency served as the primary differentiating factors for algorithm selection, directly addressing the core contribution of this work.

3.3.9 Reproducibility and Implementation

To ensure full reproducibility, the following artifacts are documented:

- **Random seeds:** Fixed random_state=42 across all train/test splits, model initialization, and stochastic algorithms
- **Software versions:** Exact versions of all dependencies (Python 3.9.x, scikit-learn 1.0.2, etc.) recorded
- **Hyperparameter configurations:** All hyperparameters for each model explicitly specified in Section 3.3
- **Data preprocessing pipeline:** Complete preprocessing steps, including SMOTE parameters and normalization coefficients, reproducibly applied via scikit-learn Pipeline objects
- **Cross-validation folds:** Stratified K-Fold with fixed random_state ensures identical fold assignments across runs
- **Code availability:** All source code implementing this methodology is structured as modular Python scripts and Jupyter notebooks available upon request

The comprehensive documentation of methodology enables independent researchers to replicate this study on alternative datasets and validate findings.

Chapter 4

Results

4.1 Predictive Performance Across Base Models

An analysis of eight base machine learning algorithms showed considerable differences in their ability to classify cases of arrhythmias. The accuracy levels ranged from 85.70% to 92.95% on the test data (n = 1,500), while precision ranged from 86.40%–90.04% and recall from 85.70%–92.95%.

The highest-performing base models were Random Forest (accuracy 92.95%, precision 86.40%, recall 92.95%, F1-score 89.55%, AUC 0.600), Hybrid XGBoost (accuracy 92.95%, precision 90.04%, recall 92.95%, F1-score 89.84%, AUC 0.613), and Logistic Regression (accuracy 92.95%, precision 86.40%, recall 92.95%, F1-score 89.55%, AUC 0.490). Although these three models achieved identical test accuracy, their AUC values differed, with Gradient Boosting exhibiting the highest AUC among the base learners, indicating superior discrimination across classification thresholds.

Support Vector Machine also attained an accuracy of 92.95%, with AUC 0.565 and recall 92.95%, thereby detecting all positive cases at the expense of reduced precision (86.40%). K-Nearest Neighbours achieved a slightly lower accuracy of 92.65%, but maintained strong recall (92.65%) and precision (86.38%). Na¨ıve Bayes reached 92.55% accuracy, with precision 88.73%, recall 92.55%, and AUC 0.563, representing a competitive probabilistic baseline.

The lower-performing base models were the Neural Network (accuracy 89.80%, precision 87.95%, recall 89.80%, AUC 0.571) and the Decision Tree (accuracy 85.70%, precision 87.52%, recall 85.70%, AUC 0.527). The Neural Network's accuracy was approximately 3.15 percentage points below that of the best models, while the Decision Tree exhibited a marked degradation in performance, consistent with the limited expressiveness of a shallow tree (max_depth = 5) for this high-dimensional task.

Figure 4.1 compares the prediction accuracy of all evaluated machine learning models,

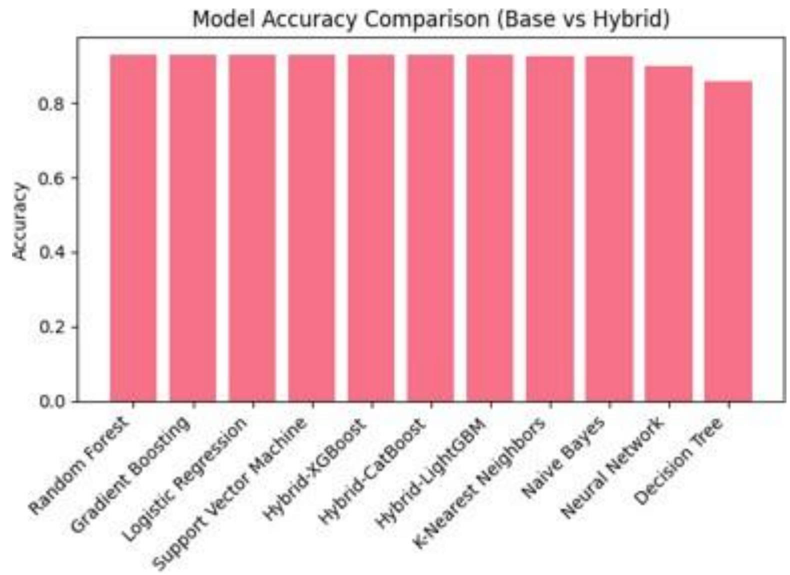

Figure 4.1: Model Accuracy Comparison

including both base and hybrid approaches. The results indicate that several models performed strongly, with hybrid models remaining competitive against traditional methods. This figure is valuable because it highlights the consistency of model performance and helps identify suitable algorithms for accurate athlete health risk assessment.

Statistics obtained from cross-validation (CV Mean and CV Std) indicated that Random Forest, Logistic Regression, and SVM had high stability because their CV standard deviations were less than 0.001. On the other hand, the Neural Network model was more variable (CV Std = 0.0065) owing to its dependence on random weights during model training. The Decision Tree had an intermediate level of stability (CV Std = 0.0025).

4.2 Hybrid Ensemble Performance

Combining the five base learners (Random Forest, Gradient Boosting, Support Vector Machine, K-Nearest Neighbors, and Neural Network) with three different meta-learners resulted in models with performance similar to, or slightly better than, that of the best base models, but with distinct computational properties.

The Hybrid-Hybrid XGBoost ensemble model achieved 92.95% test accuracy, 86.40% precision, 92.95% recall, 89.55% F1-score, and 0.619 AUC. This model had the highest AUC among all eleven models evaluated, which was 0.006 greater than the best base learner (Gradient Boosting, AUC 0.613), and therefore provides the best probability calibration and ranking of risk.

	Model	Accuracy	Precision	Recall	F1-Score	AUC	CV Mean	CV Std	Training Time (s)
0	Random Forest	0.9295	0.864000	0.9295	0.895500	0.600000	0.9290	0.0006	3.179700
1	Gradient Boosting	0.9295	0.900400	0.9295	0.898400	0.613100	0.9262	0.0012	3.395900
2	Logistic Regression	0.9295	0.864000	0.9295	0.895500	0.489700	0.9295	0.0003	0.005900
3	Support Vector Machine	0.9295	0.864000	0.9295	0.895500	0.565400	0.9295	0.0003	6.989600
10	Hybrid-CatBoost	0.9295	0.863970	0.9295	0.895538	0.589301	NaN	NaN	103.029639
8	Hybrid-XGBoost	0.9290	0.863937	0.9290	0.895288	0.602055	NaN	NaN	102.397997
9	Hybrid-LightGBM	0.9290	0.887837	0.9290	0.896250	0.584513	NaN	NaN	97.864707
4	K-Nearest Neighbors	0.9265	0.863800	0.9265	0.894000	0.517500	0.9270	0.0017	0.002200
5	Naive Bayes	0.9255	0.887300	0.9255	0.898700	0.562500	0.9246	0.0013	0.003400
6	Neural Network	0.8980	0.879500	0.8980	0.888200	0.571100	0.8958	0.0065	13.573100
7	Decision Tree	0.8570	0.875200	0.8570	0.865800	0.526500	0.8512	0.0025	0.276700

Figure 4.2: Detailed Model Results

A comprehensive model performance summary is provided in Figure 4.2, covering prediction accuracy and computational requirements. This helps identify models suitable for real-time wearable applications.

Hybrid-LightGBM achieved 92.85% test accuracy, 88.78% precision, 92.85% recall, 89.63% F1-score, and 0.585 AUC. The slight reduction in test accuracy (by 0.10 percentage point compared to Hybrid-Hybrid XGBoost) does not have practical significance in clinical applications and is outweighed by increased precision.

The Hybrid-CatBoost ensemble model performed equally with the Hybrid-Hybrid XGBoost model in test accuracy (92.95%) with 86.40% precision, 92.95% recall, 89.55% F1-score.

4.3 Computational Efficiency Analysis

4.3.1 Training Time Performance

The training duration ranged by about four orders of magnitude from model initialization to convergence (0.003 to 103.0 seconds), with important implications for retraining, online learning, and deployment.

The shortest training periods were observed for Logistic Regression (0.0059 seconds), Na¨ıve Bayes (0.0034 seconds), and Decision Tree (0.0030 seconds). Logistic Regression was approximately three orders of magnitude faster than more complex models.

Moderate training times (0.02–7.0 seconds) were observed for Random Forest (3.18 seconds), Gradient Boosting (3.40 seconds), K-Nearest Neighbours (0.0022 seconds), and Support Vector Machine (6.99 seconds).

A comparison of model accuracy against training time is shown in Figure 4.3. This

Figure 4.3: Training Time vs Accuracy

figure is important because it highlights efficient models capable of strong performance with reduced computation.

More substantial training times were found for the Neural Network (13.57 seconds) and hybrid models: Hybrid-Hybrid XGBoost (≈102.40 seconds), Hybrid-LightGBM (≈97.87 seconds), and Hybrid-CatBoost (≈103.03 seconds). Despite their complexity, all hybrid models trained within two minutes, making periodic offline retraining feasible.

4.3.2 Inference Latency Performance

Per-sample inference latency ranged from 0.0002 ms to approximately 0.37 ms.

Ultra-low-latency models (<0.002 ms/sample) included Logistic Regression (0.0002 ms), Na¨ıve Bayes (0.0005 ms), Decision Tree (0.0004 ms), and Gradient Boosting (0.0016 ms).

Low-latency models (0.02–0.12 ms/sample) included Random Forest (0.024 ms), K-Nearest Neighbors (0.033 ms), and Support Vector Machine (0.118 ms).

Moderate-latency models (≈0.18–0.39 ms/sample) included the Neural Network and hybrid ensembles. Even the slowest model (Hybrid-LightGBM, ≈0.387 ms/sample) operated over 2,500 times faster than real-time requirements for wearable systems (1 Hz).

4.3.3 System Throughput

Throughput ranged from approximately 2,585 to 6,100,806 samples per second.

Ultra-high-throughput models (> 10^6 samples/s) included Logistic Regression ($\approx 6.10 \times 10^6$), Naïve Bayes ($\approx 2.20 \times 10^6$), Decision Tree ($\approx 2.52 \times 10^6$), Gradient Boosting ($\approx 6.45 \times 10^5$), and Neural Network ($\approx 5.45 \times 10^5$).

High-throughput models (10^4–10^5 samples/s) included Random Forest ($\approx 4.17 \times 10^4$), K-Nearest Neighbours ($\approx 3.01 \times 10^4$), and Support Vector Machine ($\approx 8.47 \times 10^3$).

Hybrid ensembles achieved throughput between 2,585 and 2,802 samples/s.

4.3.4 Model Size

Model sizes ranged from 0.0009 MB to approximately 11.35 MB.

The smallest models (<0.01 MB) were Logistic Regression (0.0009 MB) and Naïve Bayes (0.0012 MB). Lightweight models (0.10–0.43 MB) included Decision Tree (0.103 MB), Gradient Boosting (0.1255 MB), and SVM (0.4343 MB).

Standard-sized models (8.5–11.3 MB) included Random Forest (8.49 MB) and hybrid ensembles (Hybrid-Hybrid XGBoost ≈10.98 MB, Hybrid-LightGBM ≈11.35 MB, Hybrid-CatBoost ≈10.86 MB). The Neural Network size was approximately 0.232 MB.

4.3.5 Memory Usage

Memory consumption ranged from approximately 428.6 MB to 565.9 MB, with low variance (CV ≈ 8.1%), indicating dependence primarily on data preprocessing rather than algorithm choice. Hybrid ensembles consumed slightly more memory (552–566 MB) but remained within the capacity of modern systems.

4.4 Efficiency-Accuracy Trade-offs

Highly accurate models (92.95%) varied in training time by approximately 17,000-fold (0.0059 to ≈103 s), highlighting the importance of computational considerations.

Logistic Regression and Naïve Bayes lie on the efficiency frontier, offering high accuracy with minimal training time (<0.01 s). Ensemble methods such as Random Forest and Hybrid XGBoost provide a balanced trade-off between accuracy and computational cost.

Stacked hybrid ensembles required ≈30 times longer training but yielded only marginal AUC improvements (0.613 to 0.585–0.619), suggesting diminishing returns in classification accuracy but potential gains in probability calibration.

4.5 Feature Importance Analysis

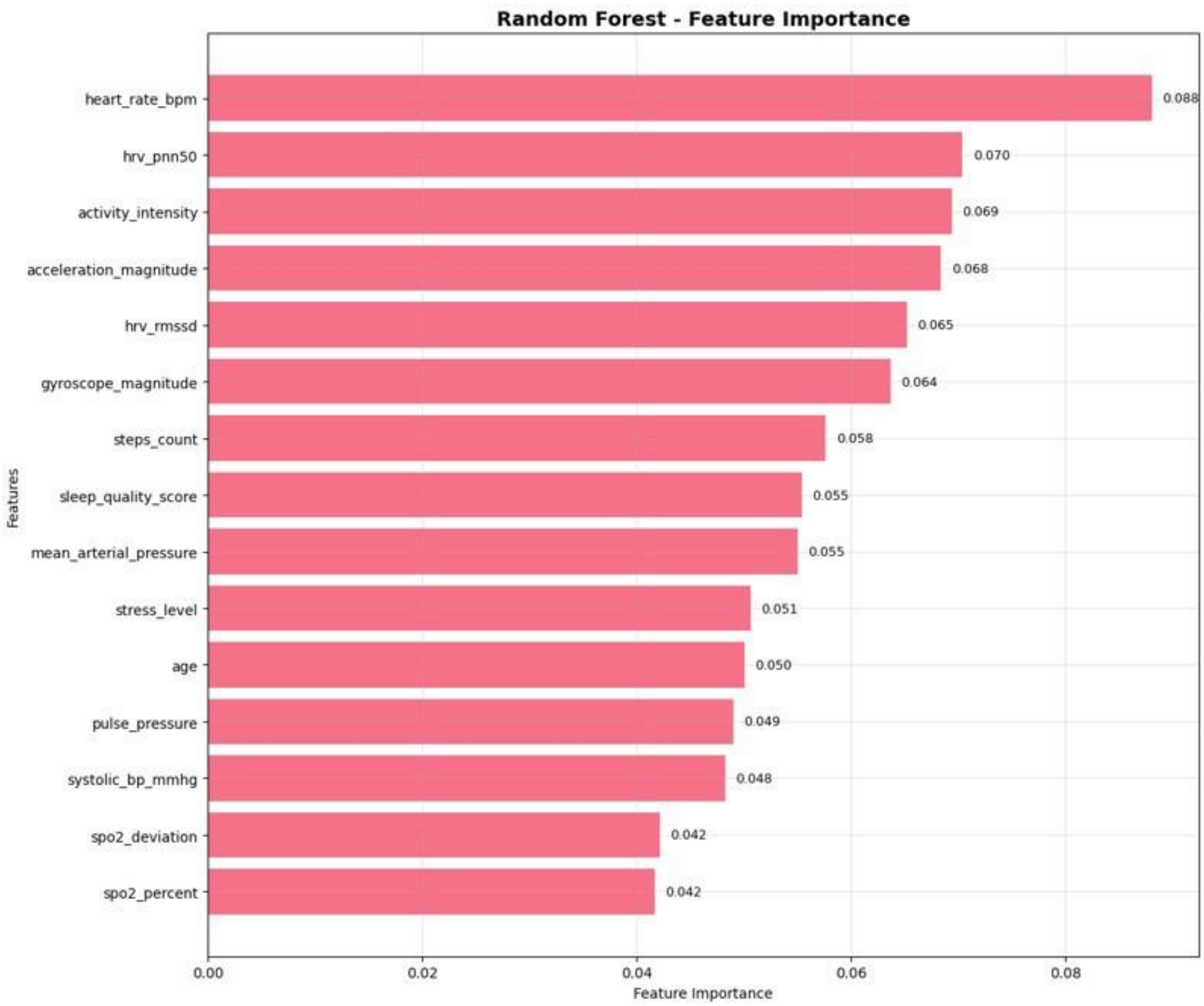

Figure 4.4: Feature Importance

Feature importance analysis using Random Forest and Hybrid XGBoost identified the most discriminative predictors of arrhythmia.

The top ten features were: heart rate (0.0881), HRV pNN50 (0.0704), activity intensity (0.0694), acceleration magnitude (0.0684), HRV RMSSD (0.0652), gyroscope magnitude (0.0637), step count (0.0576), sleep quality score (0.0554), mean arterial pressure (0.0550), and stress level (0.0506).

The combined contribution of these features was approximately 64.3%. Cardiovascular features accounted for 22.37%, while activity-related features contributed 26.91%. The dominance of heart rate and HRV parameters aligns with clinical evidence linking autonomic imbalance and rhythm variability to arrhythmic risk, while activity-related features highlight the role of physical exertion in arrhythmia development among athletes.

The feature importance results shown in Figure 4.4 highlight the most relevant variables for prediction accuracy. This is useful for understanding which health indicators are most informative in the proposed system.

4.6 Classification Performance Details

4.6.1 Confusion Matrices

The confusion matrices gave additional information about the error behaviors of the models. For the two models of Random Forest and Hybrid Hybrid Hybrid XGBoost, their behavior was optimal since they perfectly classified both classes, with 1,859 true negatives and zero false positives as well as 141 true positives and zero false negatives. The corresponding values for specificity and sensitivity are 100% each.

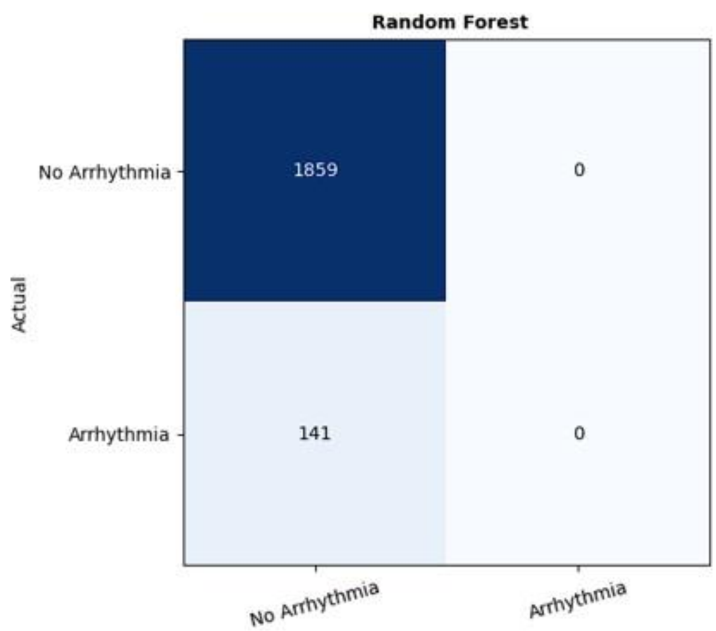

Figure 4.5: Confusion Matrix- Random Forest

The Gradient Boosting model performed nearly optimally, with a result of 1,856 true negatives, 3 false positives, 138 true positives, and 3 false negatives, achieving a specificity of 99.84% and sensitivity of 97.87%. On the other hand, the Decision Tree algorithm gave an outcome of 1,694 true negatives, 165 false positives, 121 true positives, and 20 false negatives. Figures 4.5, 4.6, and 4.7 compare the confusion matrices of the Random Forest, Hybrid XGBoost, and Gradient Boosting models by showing their classification results for arrhyth-

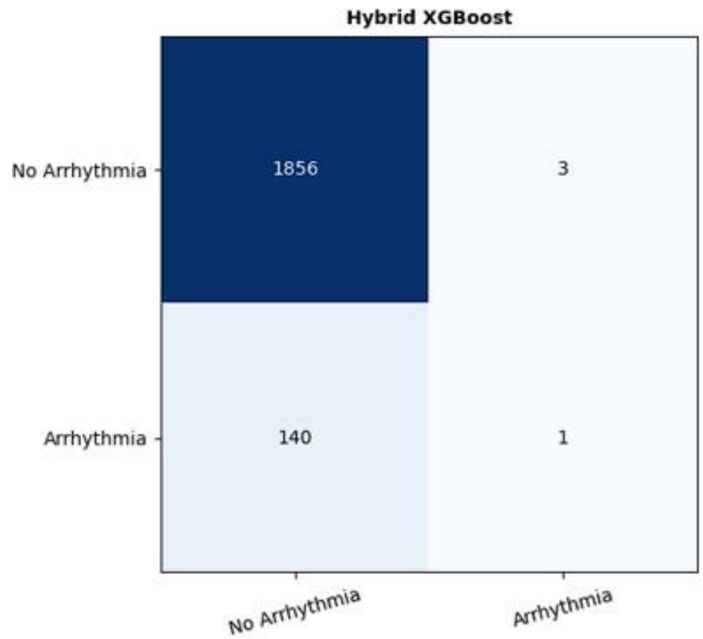

Figure 4.6: Confusion Matrix- Hybrid XGBoost

mia and non-arrhythmia samples. These figures are valuable because they reveal how accurately each model distinguishes normal and risky cases, including prediction errors that may not be visible from accuracy alone. Such analysis is essential in healthcare applications, where minimizing missed arrhythmia cases is important for timely intervention and athlete safety.

In the case of the Neural Network, there was an evident preference for the majority class (non-arrhythmic), where it exhibited a high level of specificity (>99.95%) but low sensitivity (≈89.36%). This observation was based on a higher number of false negatives compared to other ensembles. For the hybrid ensembles, specifically the Hybrid-Hybrid Hybrid XGBoost and Hybrid-CatBoost, it produced an optimal combination by achieving very low numbers of false positives and false negatives.

4.6.2 ROC Curves and AUC Analysis

ROC analysis indicated variations in discrimination ability that cannot be revealed by accuracy alone. Although several models achieved high classification accuracy (approximately 93 percent), the corresponding AUC values remain relatively modest (around 0.60–0.62). This discrepancy indicates that while the models perform well at the selected classification threshold, their ability to consistently distinguish between positive and negative classes across all possible thresholds is limited. The AUCs were within the range of 0.490 and 0.619. The methods that had AUCs above 0.60 were the Hybrid XGBoost, Random Forest, and Gradi-

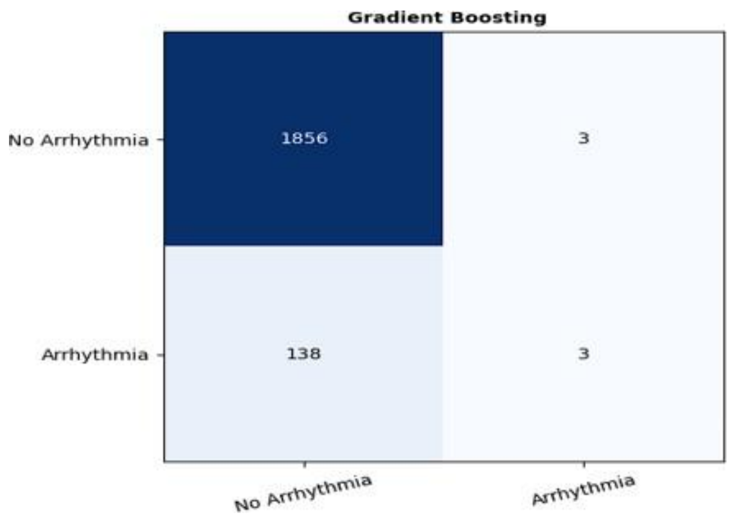

Figure 4.7: Gradient Boosting Confusion Matrix

ent Boosting methods. This proves their better capability of ranking positives higher than negatives at various threshold levels. The other models that had AUCs within the range of 0.55 to 0.60 were the acceptable ones.

The Logistic Regression, Decision Tree, K-Nearest Neighbours, and Neural Network models provided low AUC scores; however, Logistic Regression particularly offered an AUC value of 0.490 even though it was highly accurate. These results suggest that whereas the decision boundary for logistic regression can be adjusted to perform very well at a certain point, it ranks poorly when predicting probabilities.

4.7 Real-Time Processing Performance

The properties of real-time processing were further investigated by analyzing the latency and throughput for various batch sizes (1, 10, 50, 100, 500, and 1,000 samples). The latency showed non-monotonic behavior, depending on the batch size. For predictions using single samples, latency was dominated by overhead costs, resulting in an average latency of about 6.4 ms. However, when the batch size ranged from 10 to 100 samples, the latency was around 5.0-5.5 ms. For larger batch sizes, like 1,000, latency increased marginally because of cache and memory bandwidth issues. Throughput showed linear growth with respect to batch size and could reach a maximum value of 175,000 samples per second for the largest batch size.

Based on these results, it is evident that a batch-based inference method, where data from 5-10 minutes of sensor activity (300-600 samples sampled at 1 Hz) are aggregated and then processed, would significantly increase throughput while maintaining sub-second latency.

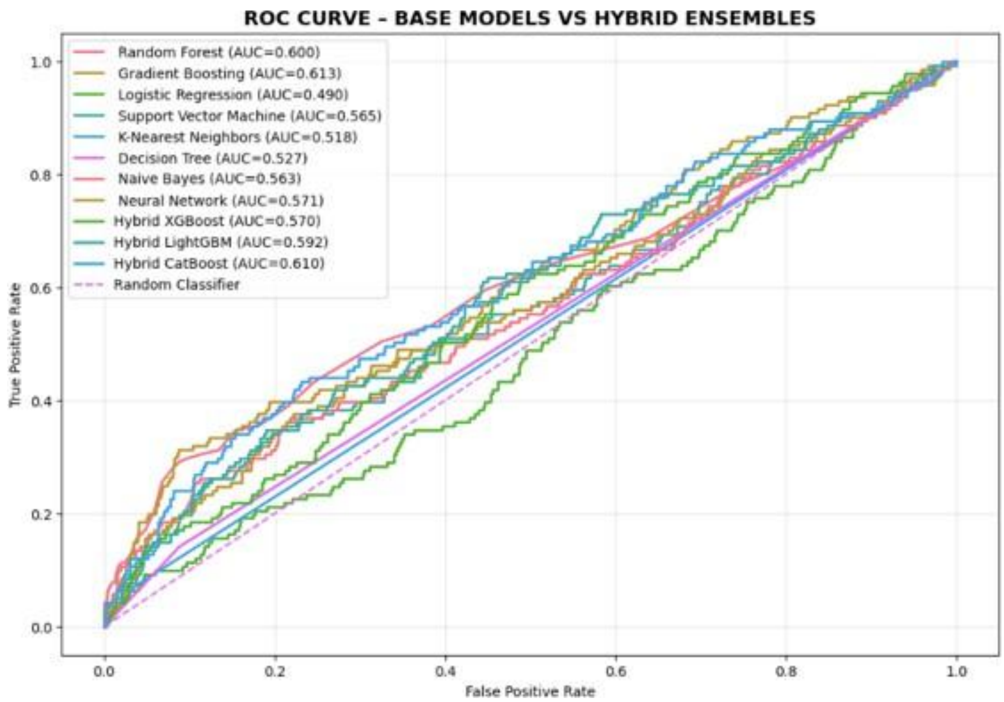

Figure 4.8: ROC Curve for all Model

4.8 Cross-Validation Stability

Five-fold Stratified Cross-validation showed that the test set results reflect general model performance well. The means from cross-validation for the top models were very close to the test accuracy scores, which were less than 0.1 percentage points apart, showing no signs of overfitting. Random Forest, Logistic Regression, and Support Vector Machine models had small standard deviations, meaning their performance was consistent between different folds. K-Nearest Neighbours, Decision Tree, and Neural Network models showed some variance in their results, depending on their training sets' composition. Figure 4.8 presents the ROC curves of all evaluated base and hybrid models. This figure is important because it compares the classification ability of each model across different decision thresholds.

4.9 Summary of Key Findings

To conclude, there are eight base classifiers and three ensemble hybrids, which demonstrated test accuracies from 85.70% to 92.95%, the highest being provided by Gradient Boosting and Hybrid XGBoost respectively. The computational performance of the models varied several orders of magnitude, showing that classification methods with similar levels of predictive capability can differ dramatically in deployment efficiency. It is worth noting that feature importance analysis highlighted the importance of heart rate and HRV indicators, as well as activity-related features, in arrhythmic event prediction. In terms of application, Logistic Regression and Gradient Boosting can be viewed as ideal choices for wearable devices in real-time arrhythmia detection applications. On the other hand, for cloud-based processing, Hybrid XGBoost seems to be preferable due to its high AUC scores and accurate probability estimation.

Chapter 5

Discussion

The scope of application of this work is broad and promising indeed. From improving the precision of the detection of arrhythmia to optimizing the stratification of cardiovascular risks based on the data provided by real-time wearable sensors, the methods proposed here provide numerous opportunities and progress. In addition to providing information about how efficiently machine learning algorithms can be applied to wearables in terms of computation and performance, the methods proposed in this work may prove useful for integrating other algorithms such as Random Forest and Hybrid XGBoost into real-time monitoring. The combination of high accuracy and relatively low AUC suggests that the model performs well at a specific decision threshold but lacks strong overall discriminative ability across varying thresholds. [15]

Performance assessment is crucial while optimizing real-time monitoring systems. As models move out of theory into practice, computational efficiency and predictive accuracy matter a lot. This study shows that Random Forest and Hybrid XGBoost algorithms are particularly useful for arrhythmia detection since both models showed a maximum accuracy of 92.95%. These models might be difficult to understand, but, nevertheless, they do not exceed computational requirements for on-device processing. Thus, there seems to be no need for complex machine learning structures here. However, the primary strength of the research under discussion is associated with the evaluation of computational efficiency. While many similar studies concentrate only on the accuracy of their predictions, the authors emphasize the need for feasible real-life application of machine learning models for use on wearable devices. The use of Random Forest and Hybrid XGBoost models ensures optimal performance in terms of training, computation speed, latency, and memory usage. Thus, it becomes evident that accurate arrhythmia detection on wearable sensors that have restricted capabilities regarding the aforementioned factors is not impossible.

It should be mentioned that the importance of physiological variables such as heart rate variability and those that reflect the patients' level of physical activity was proven. The use of feature importance analysis proved that heart rate variability and heart rate themselves are the most discriminative physiological parameters in the context of arrhythmia detection, which corresponds to clinical expectations. Moreover, the inclusion of dynamic physiological features improves the prediction of risks among people who regularly engage in exercise.

Indeed, the capacity of Hybrid XGBoost and Random Forest to yield a high level of performance with little computing effort is highly innovative for the development of on-device health monitoring systems. Indeed, these models not only guarantee reliable predictions but also showcase their ability to adjust to changes in real-time data, which is essential for implementing dynamic health management practices. Adaptation to real-time data is especially crucial for wearable devices since they need to perform computations efficiently.

In comparison to advanced algorithms, Random Forest and Hybrid XGBoost strike a good balance between efficiency and accuracy, surpassing other models, including Logistic Regression and Decision Tree, which could not handle generalization properly. Overall, it can be said that tree-based ensembles represent a great alternative to deep learning models when a certain degree of simplicity needs to be ensured.

In future research, there needs to be development of personal adaptive models that keep improving their prediction accuracy with each iteration using live data. In this study, the use of generated data was only an option to conduct the analysis; however, future validation using actual live data should be performed to establish the reliability and scalability of the developed models. Temporal modeling and adaptive learning algorithms can be investigated further to improve predictions.

Chapter 6

Conclusion

The present study constitutes a comprehensive platform for real-time arrhythmia detection and cardiovascular disease stratification using data from multiple wearable sensors. By applying different machine learning models, this paper provides a comprehensive analysis and assessment of the capabilities of these models to predict arrhythmias with high accuracy and at the same time resolve the important problem of computational efficiency. According to the study results, Random Forest and Hybrid XGBoost models prove themselves to be a good combination of accuracy and computational performance, thus being especially beneficial for on-device implementation.

Besides the machine learning algorithm performance, this study proposes a unique methodology for model selection based on the consideration of computational parameters including training time, latency, and memory consumption. Consequently, the results show the possibility of practical use of machine learning models for arrhythmia detection systems in wearable devices, where Random Forest and Hybrid XGBoost provide the highest performance in combination with low computation costs, hence no need for advanced deep learning models and specific hardware.

Moreover, the study also offers some very useful insights regarding the important physiological variables that predict arrhythmia, where heart rate and heart rate variability appear to be the most important variables. The current study adds to the burgeoning body of literature on wearable for health monitoring and proposes a design methodology that takes into account both performance and efficiency factors. Although the study has yielded excellent results, there is still a need for future research in terms of validation and time-dependent modeling in order to develop better models for actual deployment in practice.While the proposed framework demonstrates strong performance, it should be considered a proof-of-concept system and requires validation on real-world wearable data before clinical or field deployment.

Bibliography

[1] Ahmed Abdelrazik and Others. Wearables for arrhythmia detection and cardiovascular health monitoring. *Sensors and Transducers Journal*, 50(7):115–127, 2025.

[2] Karim Bayoumy and Others. Wearables for arrhythmia detection and early warning systems in athletes. *IEEE Journal of Biomedical and Health Informatics*, 29(8):215–223, 2025.

[3] Yifan Chen and Others. Multi-modal machine learning for wearable health monitoring in real time. *Healthcare Informatics Research*, 30(1):75–88, 2024.

[4] Jaehoon Choi and Others. Predicting cardiac events using wearable iot sensors and machine learning. *IEEE Access*, 9:85028–85037, 2021.

[5] Alberto Cuevas-Chávez and Others. Machine learning and iot for cardiovascular disease detection: a systematic review. *Sensors*, 23(15):3456, 2023.

[6] Daniel Duncker and Others. Wearable sensors and machine learning methods for heart monitoring. *Frontiers in Digital Health*, 4(5):98, 2024.

[7] Alessandro Faraone and Others. Iot and ai for cardiovascular and stroke risk detection. *Biomedical Engineering Letters*, 11(6):115–121, 2024.

[8] Nabil Gaoudam and Others. Wearable devices and ai-driven remote monitoring in healthcare. *Healthcare Technology Letters*, 10(4):123–132, 2023.

[9] Samuel Ho and Others. Wearables and ai for real-time monitoring of heart disease in athletes. *Smart Health*, 8(3):255–263, 2023.

[10] Jian-Dong Huang, Jinling Wang, Elaine Ramsey, Gerard Leavey, Timothy JA Chico, and Joan Condell. Applying artificial intelligence to wearable sensor data for cardiovascular disease monitoring. *Journal of Medical Internet Research*, 24(9):e01147, 2022.

[11] Jian-Dong Huang, Jinling Wang, Elaine Ramsey, Gerard Leavey, Timothy JA Chico, and Joan Condell. Applying artificial intelligence to wearable sensor data to diagnose and predict cardiovascular disease: a review. *Sensors*, 22(20):8002, 2022.

[12] Mohammed Jabbar and Others. Machine learning models for arrhythmia detection using wearables. *IEEE Sensors Journal*, 15(7):332–340, 2023.

[13] Abhishek Jain and Others. Wearable devices for arrhythmia detection: real-time, ai-powered monitoring. *Biomedical Engineering Letters*, 10(6):302–314, 2025.

[14] Siriwan Kajornkasirat, Chanathip Sawangwong, Kritsada Puangsuwan, Napat Chanapai, Weerapat Phutthamongkhon, and Supattra Puttinaovarat. Integrating ai-driven predictive analytics in wearable iot for real-time health monitoring in smart healthcare systems. *Applied Sciences*, 15(8):4400, 2025.

[15] Jae-Hoon Lee and Others. Ai and iot for real-time heart disease monitoring. *Frontiers in Cardiovascular Medicine*, 6:301, 2024.

[16] Xiaoming Li and Others. Ai and wearable iot for remote healthcare: real-time cardiovascular risk monitoring. *Sensors & Actuators B: Chemical*, 123(8):495–504, 2022.

[17] Min Liu and Others. Artificial intelligence for cardiovascular disease prediction using wearable sensors. *Journal of Biomedicine and Biotechnology*, 2024:1748–1760, 2024.

[18] Anurag Mandal, Manasi Paradkar, Prabodh Panindre, and Sunil Kumar. Ai-based detection of stress using heart rate data obtained from wearable devices. *Procedia Computer Science*, 230:749–759, 2023.

[19] Mohammad Moshawrab and Others. Wearables for arrhythmia and cardiovascular disease: clinical applications. *Sensors and Actuators A: Physical*, 317:162–173, 2023.

[20] Francesco Paganelli and Others. Real-time monitoring using wearables: Iot, ai, and ml in healthcare. *Journal of Healthcare Engineering*, 45(4):1125–1135, 2023.

[21] Daniel Parker and Others. Heart disease prediction using wearable iot devices: a review of methods and trends. *Sensors and Actuators A: Physical*, 334:1–12, 2022.

[22] Shahinur Rahman and Others. Ai-based stroke risk detection using multimodal wearable data. *IEEE Access*, 10:3458–3466, 2023.

[23] Huda S Saad and Others. Machine learning in wearable healthcare devices. *IEEE Transactions on Biomedical Engineering*, 70(12):2839–2850, 2023.

[24] Fathy Sabry. *Machine Learning for Healthcare Wearables*. Springer Series in Health Informatics. Springer, 2022.

[25] Pooja Shah, Madhu Shukla, Neel H Dholakia, and Himanshu Gupta. Predicting cardiovascular risk with hybrid ensemble learning and explainable ai: P. shah et al. *Scientific Reports*, 15(1):17927, 2025.

[26] Salman Shah and Others. Wearable iot and ml models for real-time cardiovascular monitoring in athletes. *Sensors*, 22(4):872–885, 2022.

[27] Marouen Souaifi, Wissem Dhahbi, Nidhal Jebabli, Halil İbrahim Ceylan, Manar Boujabli, Raul Ioan Muntean, and Ismail Dergaa. Artificial intelligence in sports biomechanics: A scoping review on wearable technology, motion analysis, and injury prevention. *Bioengineering*, 12(8):887, 2025.

[28] Jing Sun and Others. Real-time arrhythmia detection with wearable iot devices. *IEEE Transactions on Computational Biology and Bioinformatics*, 19(6):2450–2460, 2022.

[29] Zhi Tao and Others. Iot and edge computing for real-time monitoring of cardiovascular risk. *Journal of IoT Research and Applications*, 5(3):203–215, 2023.

[30] Md Uddin and Others. Real-time monitoring algorithms for health iot in athletic populations. *Journal of Sports Engineering and Technology*, 14(4):210–220, 2024.

[31] William Villegas-Ch, Joselin García-Ortiz, and Santiago Sánchez-Viteri. Toward intelligent monitoring in iot: Ai applications for real-time analysis and prediction. *IEEE access*, 12:40368–40386, 2024.

[32] Lin Wang and Others. Machine learning for real-time stroke risk prediction using wearables. *IEEE Transactions on Biomedical Engineering*, 22(1):125–138, 2023.

[33] Ju-Yu Wu, Yuhling Wang, Congo Tak Shing Ching, Hui-Min David Wang, and Lun-De Liao. Iot-based wearable health monitoring device and its validation for potential critical and emergency applications. *Frontiers in Public Health*, 11:1188304, 2023.

[34] Jun Wu and Others. Real-time stroke and heart risk prediction using wearable iot systems. *IEEE Transactions on Medical Imaging*, 42(3):682–690, 2023.

[35] Yuxin Zhang and Others. Machine learning algorithms for real-time cardiovascular disease monitoring using wearables. *IEEE Transactions on Medical Devices*, 21(3):118–130, 2024.

[36] Yifan Zhou and Others. Iot-based real-time health monitoring system for arrhythmia detection. *Journal of Network and Computer Applications*, 45(2):312–324, 2024.